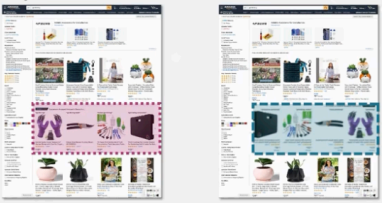


Probability and Statistics

Wednesday, 14 February 2024 06:25

Hypothesis Testing (AB Testing)

- Would it be better to have a row of other recommendations instead of "Amazon's Choice"?
- How can you decide?



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I must be careful that users going to each option don't have BIAS
a specific region, products not seasonal in one and non-seasonal in the other

We will compare the centers of each distribution to see which strategy is more successful

Hypothesis Testing (AB Testing)

- Would it be better to have a row of other recommendations instead of "Amazon's Choice"?
- How can you decide?
- Put another version of the site in production and see which version has a higher rate of visitors purchasing



% of visitors purchasing through previous version

% of visitors purchasing through new version

2.1

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RANDOM VARIABLES

Random Variables

- A random variable X contains a set of possible outcomes for a random event
 - Dice roll: $X \in \{1,2,3,4,5,6\}$
 - Catching ball: $X \in \{Yes, No, Almost\}$
 - Next word in sentence: $X \in \{alligator, bike, pony, grass, bird, tree, \dots\}$
 - Coin Flip $X \in \{heads, tails\}$
 - It's usually better to represent the outcome as a number $X \in \{0,1\}$



She ran outside to ride her

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Random vs non-random variables

- Random
 - How many gears a car has
 - Roll of the dice
- Not random
 - How many gears your car has
 - Previous dice roll

- What makes a variable random vs non-random?
 - Having a set of probable outcomes

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Random vs non-random variables

- Random
- Not random

» How many gears **a** car has

	Probability
3 on the tree	Very low
4 on the floor	Low
5-speed	High
6-speed+	Moderate

» How many gears **your** car has

	Probability
3 on the tree	0%
4 on the floor	0%
5-speed	100%
6-speed+	0%

• What makes a variable random vs non-random?

» Having a set of **probable** outcomes

Probability distribution

PROBABILITY DISTRIBUTIONS

Distributions

• Probability distribution of a random variable

» How the probability of the random variable's outcomes is spread out across the outcomes

How many gears a car has:

Gears	3	4	5	6+
Probability	Very Low	Low	High	Moderate

Distributions

• Probability distribution of a random variable

» How the probability of the random variable's outcomes is spread out across the outcomes

Roll of a dice:

Side	1	2	3	4	5	6
Probability	1/6	1/6	1/6	1/6	1/6	1/6

Distributions

• Probability distribution of a random variable

» How the probability of the random variable's outcomes is spread out across the outcomes

She ran outside to ride her _____.

Next word	alligator	bike	pony	grass	Bike or pony
Probability	Very low	Very high	Very low	Almost zero	Super high

Distribution Types

• Discrete

- » Child/Adult
- » Short/Average/Tall
- » Did they buy something from Amazon.com?
- » Did you vote this year?
- » Months in the year

• Continuous

- » Age
- » Height
- » Amount spent at Amazon.com
- » Average number of people that voted

Distribution Types

• A **Discrete** variable maintains its output, or has no output, across specific distances of its x-axis

» A graph of a discrete variable's outcomes has "bars".

Roll of a dice:

Side	1	2	3	4	5	6
Probability	1/6	1/6	1/6	1/6	1/6	1/6

Distribution Types

• A **Continuous** variable can take any real value

Discrete Distributions

• **Probability mass function (pmf)**

» The distribution of probabilities for a discrete random variable

• If X is a discrete random variable, its pmf is a function is

» $f: \Omega \rightarrow \mathbb{R}$ such that for all $x \in \Omega$:

$$1 \geq f(x) \geq 0$$

$$\sum_{x \in \Omega} f(x) = 1$$

x	f(x)
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

Handwritten notes: $1/6 + 1/6 + 1/6 + 1/6 + 1/6 + 1/6 = 1$

Continuous Distributions

• A **probability density function (pdf)** is the distribution of

- » Independent or what's around it
- » A graph of a continuous variable is "smooth"



probabilities for a continuous random variable

- If X is a continuous random variable, its pdf is an (integrable) function $f: \mathbb{R} \rightarrow \mathbb{R}$ for all $x \in \mathbb{R}$:

$$1 \geq f(x) \geq 0$$

$$\int_{\mathbb{R}} f(x) dx = 1$$



DISCRETE DISTRIBUTION

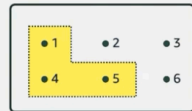
Probability Diagram

- A **probability space** contains possible outcomes
- A **random variable** holds possible outcomes
- An **event** is a collection of random variables

$$X \in \{1, 2, 3, 4, 5, 6\}$$

$$E \in \{1, 4, 5\}$$

$$P(X \in \{1, 4, 5\}) = \frac{1}{2}$$



$$P_X(E) = \sum_{x \in E} f(x)$$

$$P(X \in \{1, 4, 5\}) = P(X \in \{1\}) + P(X \in \{4\}) + P(X \in \{5\}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

Independence

- A collection of random variables are independent if the outcome of one does not affect the outcome of any others
 - » Coin flips **are independent** since the outcome of the first flip does not affect the outcome of the second
 - » The probability of being 6ft tall is **NOT independent** of the probability a parent being 6ft tall
 - » The card dealt while playing Blackjack is **NOT independent** after the first card has been dealt

$$P(A \cap B) = P(A)P(B)$$

joint probability of A and B

Discrete Distributions: Uniform



Discrete Distributions: Bernoulli

- Bernoulli Distribution
 - » A distribution that will always have 2 possible values $\{0, 1\}$
 - » A random variable with this distribution can end up being 1 with a probability p , and therefore 0 with a probability $1 - p$

$$f(k; p) = \begin{cases} p & k = 1 \\ 1 - p & k = 0 \end{cases}$$

Did an Amazon.com visitor purchase?

p	.1%
$1 - p$	99.9%



Bernoulli Distribution

- » The number of successes within a series of events where each event on its own has a Bernoulli Distribution
- » A count of the successes from a series of Bernoulli distributed events

$$f(k; n, p) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Note: The notation $\binom{n}{k}$ comes from Counting Theory. This formula can be derived from Counting Theory principles.

Control: existing solution Test: new solution

Discrete Distributions: Bernoulli

- Bernoulli Distribution
 - » The number of successes within a series of events where each event on its own has a Bernoulli Distribution
 - » A count of the successes from a series of Bernoulli distributed events

» A count of the successes from a series of Bernoulli distributed events

$$f(k; n, p) = \binom{n}{k} p^k (1-p)^{n-k}$$

How many Amazon.com visitors purchased on the same day?

Control	???
Test	???



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Discrete Distributions: Geometric

- The **Geometric Distribution** represents the number of tries needed before the first success
 - How many tries are expected before success

$$f(k; p) = (1-p)^{k-1} p$$

How many Amazon.com visitors visit the site before one of them purchases a product?

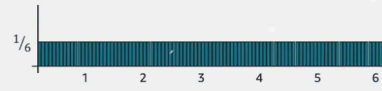
???



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From Discrete to Continuous

- Discrete distributions can approximate continuous distributions by creating smaller and smaller discrete "buckets"



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Cannot calculate it with distributions with discontinuities

Integration

- Integration can find the area under almost any function between any two values



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Integration gives us the ability to find the area between those discontinuous points we have approximated to infinity

From Discrete to Continuous

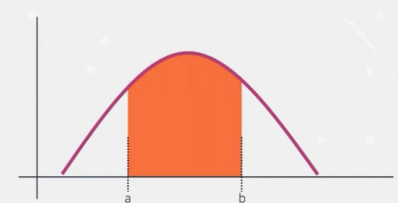
- Discrete distributions can approximate continuous distributions by creating smaller and smaller discrete "buckets"
 - When the buckets get infinitely small, it is the same as a continuous distribution
 - This is called **integration**!



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It can be reduced to a couple of values

- Integration can find the area under almost any function between any two values



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CONTINUOUS DISTRIBUTIONS

Continuous Distributions

- A probability density function (pdf) is the distribution of probabilities for a continuous random variable
- If X is a continuous random variable, its pdf is an (integrable) function $f: \mathbb{R} \rightarrow \mathbb{R}$ for all $x \in \mathbb{R}$:

$1 \geq f(x) \geq 0$
 $\int_{\mathbb{R}} f(x) dx = 1$

$1 \geq f(x) \geq 0$
 $\sum f(x) = 1$
 PMF (disc)



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Continuous Distributions: Uniform

- Uniform Distribution**
 - Represents a constant value for the pdf

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Continuous vs Discrete Distributions

- Continuous**


$$P_X([a, b]) = \int_a^b f(x) dx$$
- Discrete**

Gears	3	4	5	6+
Probability	Very Low	Low	High	Moderate

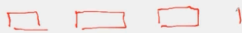


$$P_X(E) = \sum_{x \in E} f(x)$$

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- » The probability that a random variable is between a and b is a multiple of that constant value

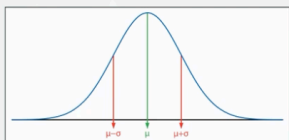
$$f(a, b) = c(b - a)$$



Continuous Distributions: Gaussian

- Gaussian Distribution
 - » One of the most common distributions
 - » Also called the Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

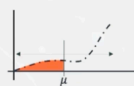


DESCRIBING DISTRIBUTIONS

Integration

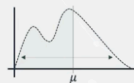
- There are 2 main characteristics of distributions

- » Center
- » Mean



- » Spread

- » Variance
- » Standard Deviation
- » $Var = std^2$
- » $\sqrt{Var} = std$



Expected values

- The expected value of a random variable is the most likely value to be randomly selected from the distribution
 - » Pick a number at random, what value is expected?
- The Expected Value is a weighted average of distribution values multiplied by their respective probability:

$$E[X] = \sum x f(x)$$

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

Expected values by distribution type

- Discrete distribution
 - » Mean

$$\sum x f(x)$$

$$E[g(x)] = \sum g(x) f(x)$$

- » Standard deviation

$$\sqrt{\sum (x - \mu)^2 f(x)}$$

$$\int_{-\infty}^{\infty} g(x) f(x) dx$$

$$Var = E[X^2] - E[X]^2$$

Expected values by distribution type

- Continuous distribution
 - » Mean

$$\int_{-\infty}^{\infty} x f(x) dx$$

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

- » Standard deviation

$$\sqrt{\int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx}$$

$$\int_{-\infty}^{\infty} g(x) f(x) dx$$

$$Var = E[X^2] - E[X]^2$$

HYPOTHESIS TESTING

Hypothesis Testing (AB Testing)

- Model a user's chance of clicking as a Bernoulli distribution
 - » n is the chance of success (user)



purchases)

- The goal is to estimate p for both the control and test
 - Users who see current vs new recommendations

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Hypothesis Testing (AB Testing)

- Put both of sites in production and get a dataset
- What's the **best metric** to use for $\hat{p}_c$ and $\hat{p}_t$?
 - For hypothesis testing

$\hat{p}_c$		$\hat{p}_t$	
Visitor Tag	Purchased product	Visitor Tag	Purchased product
849000774	0	849000774	0
84223934	1	84223934	0
83202084	0	83202084	0
84542229	1	84542229	1
8339352	0	8339352	0
84312122	0	84312122	0
84423202	1	84423202	0
84111776	0	84111776	1
845442	0	845442	1
833443	1	833443	0
8443232	0	8443232	0
4888763	1	4888763	0
8350857	1	8350857	0
8400052	0	8400052	0
84228225	1	84228225	0
8345421	0	8345421	0
84567549	1	84567549	0
83202048	0	83202048	0
84423545	1	84423545	1
83103548	1	83103548	0
8332354	0	8332354	1
84567534	0	84567534	0
83204556	0	83204556	1
84523542	0	84523542	0
8324563	1	8324563	0
8445367	0	8445367	0
83242675	0	83242675	0
8346245	1	8346245	1
84592028	0	84592028	0

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Hypothesis Testing (AB Testing)

- Think about each user outcome as a random variable X_i
 - The average value:

$$\frac{X_1 + X_2 + \dots + X_n}{n}$$
 - is a good estimate for p

$$\hat{p}_i = \frac{X_1 + X_2 + \dots + X_{n_i}}{n_i}$$

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Hypothesis Testing (AB Testing)

- Central Limit Theorem
 - The average of many independent random variables with the same distribution is approximately a normal distribution

$$\hat{p}_i = \frac{X_1 + X_2 + \dots + X_{n_i}}{n_i} \sim N\left(p_i, \sqrt{\frac{p_i(1-p_i)}{n_i}}\right)$$

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Hypothesis Testing: The Null Hypothesis

- The **Null Hypothesis** is a hypothesis that you work to reject:

$$\hat{p}_{control} = \hat{p}_{test}$$

Hypothesis Testing: The Null Hypothesis

- Shopping example:
 - Try to prove there's a difference in these purchase rates

- » Difficult
 - » Easier to prove the opposite is NOT true
 - » Prove the purchase rates are not the same
- ❖ "We failed to reject the null hypothesis"
- » The test has an unclear result
 - » Cannot prove the hypothesis, instead fail to reject the null hypothesis

Hypothesis Testing (AB Testing)

❖ Steps

- » New random variable would have:

$$N(0, \sqrt{\hat{\sigma}_c^2 + \hat{\sigma}_t^2})$$

- » Divide by the variance of the new random variable
- » End up with a distribution that is $N(0,1)$

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Hypothesis Testing (AB Testing)

❖ Z-score

- » How far a data point is from the mean
- » Calculated for every value on the x-axis, relative to the area under a $N(0,1)$
- » Only need to compare the transformed numbers to the Z-scores to find out if a difference is "significant"

$$??? = \frac{\hat{p}_c - \hat{p}_t}{\sqrt{\hat{\sigma}_c^2 + \hat{\sigma}_t^2}}$$

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Hypothesis Testing (AB Testing)

❖ "One-tail" test

- » Look at confidence of one side
- » Only when the test value can only be higher or lower than the control

❖ Example

- » The height of a child will only go up
- » Testing if a child is shorter than before doesn't make sense

$$1.65 \leq \frac{\hat{p}_c - \hat{p}_t}{\sqrt{\hat{\sigma}_c^2 + \hat{\sigma}_t^2}}$$

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PAGE RANK

This is the way Google used since 1998 to establish a ranking of the most relevant pages
It is based on 2 techniques: the law of probability and Eigen Vectors
is to analyze web page links and assign each one a global probability



We would start with a Ranking Vector with equal probabilities

We would start with a Ranking Vector with equal probabilities

$$\mathbf{r} = [r_A, r_B, r_C, r_D, r_E]^T = [0.2, 0.2, 0.2, 0.2, 0.2]^T$$

If we analyze link dependencies, we see from visits that if we are at point D, 100% of the time we go to A, so its probability is transferred

$$r_A = r_D.$$

For B, it depends on those from A and E; those from A can go to 3 different links, while those from E only go to B

$$r_B = \frac{r_A}{3} + r_E.$$

Following this probability mechanism we calculate the rest

$$r_C = \frac{r_B}{2}, \quad r_D = \frac{r_A}{3}, \quad \text{and} \quad r_E = \frac{r_A}{3} + \frac{r_B}{2} + r_C.$$

In the end what we have is a probability matrix

It is called a stochastic matrix and all its columns sum to 1

$$\mathbf{L} = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 1/3 & 0 & 0 & 0 & 1 \\ 0 & 1/2 & 0 & 0 & 0 \\ 1/3 & 0 & 0 & 0 & 0 \\ 1/3 & 1/2 & 1 & 0 & 0 \end{pmatrix},$$

A Ranking Vector is nothing other than an eigen vector of a matrix with eigen value = 1
calculate it and sort products by importance

```
L = np.array([
    [0, 0, 0, 1, 0],
    [1 / 3, 0, 0, 0, 1],
    [0, 1 / 2, 0, 0, 0],
    [1 / 3, 0, 0, 0, 0],
    [1 / 3, 1 / 2, 1, 0, 0],
])

print("Link matrix: \n{L}\n")

eigenvalues, eigenvectors = np.linalg.eig(L) # Gets the eigenvalues and vectors
idx = eigenvalues.argsort()[::-1] # Sort by the eigenvalues (we know the largest is 1)
largest = (
    np.array(eigenvectors[:, idx]).flatten().real
) # Select the eigenvector corresponding to largest eigenvalue
ranking_eig = np.abs(
    largest / np.linalg.norm(largest, 1)
) # normalise the eigenvector -- this is the ranking vector!

print(f"Ranking vector (by eigendecomposition):\n{ranking_eig}\n")
print(f"---- the sum of the updated ranking vector:\n{np.sum(ranking_eig)}")
```

```
Link matrix:
{L}

Ranking vector (by eigendecomposition):
[9.00719561e-17 4.00000000e-01 2.00000000e-01 4.47422882e-17
 4.00000000e-01]

---- the sum of the updated ranking vector:
1.0
```

Using this ranking vector, we can order the five product in the order of their importance: E, B, C, A, and D.

When not all links point to each other, or we have a much larger dataset, we must operate differently

Having a product only accessible from E
This causes the Eigen Vector to come out wrong

```
Ranking vector (by eigendecomposition):
[0. 0. 0. 0. 0. 1.]

---- the sum of the updated ranking vector:
1.0
```

One proposed solution is a damping factor of .85 that at least links all unconnected products, forming what is called a Google Matrix

```
Link matrix:
[[0. 0. 0. 1. 0. 0.]
 [0.33333333 0. 0. 0.5 0. ]
 [0. 0.5 0. 0. 0. 0.]
 [0.33333333 0. 0. 0. 0. 0.]
 [0.33333333 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 1.]
```

```
[0.33333333 0. 0. 0. 0. 0. 0. 0. ]
[0.33333333 0.5 1. 0. 0. 0. 0. 0. ]
[0. 0. 0. 0. 0.5 1. 0. 0. ]

Reformulated link matrix:
[0.025 0.025 0.025 0.875 0.025 0.025 0.025 0.025 ]
[0.30833333 0.025 0.025 0.025 0.45 0.025 0.025 0.025 ]
[0.025 0.45 0.025 0.025 0.025 0.025 0.025 0.025 ]
[0.30833333 0.025 0.025 0.025 0.025 0.025 0.025 0.025 ]
[0.30833333 0.45 0.875 0.025 0.025 0.025 0.025 0.025 ]
[0.025 0.025 0.025 0.025 0.45 0.875 0.025 0.025 ]

Ranking vector (by eigendecomposition):
[0.06092206 0.10400869 0.06920369 0.04226125 0.14528808 0.57831623]

└─ the sum of the updated ranking vector:
0.9999999999999999
```

With a very large dataset of billions of links, PowerIteration can be used using the Google matrix, iterating by multiplying until finding the eigen vector with eigen value of 1 approximately rather than exactly

```
v = np.array([[0.2], [0.2], [0.2], [0.2], [0.2], [0.2]])
print("Initial ranking vector:\n", v.T)

L = np.array([
    [
        [0, 0, 0, 1, 0],
        [1 / 3, 0, 0, 0, 1],
        [0, 1 / 2, 0, 0, 0],
        [1 / 3, 0, 0, 0, 0],
        [1 / 3, 1 / 2, 1, 0, 0],
    ]
])

# print('Link matrix:\n', L)

d = 0.85
N = L.shape[1]
L_d = d * L + (1 - d) / N
print(f"Reformulated link matrix:\n{L_d}\n")

v = L_d.dot(v)
print(f"Update of the ranking vector:\n{v.T}\n")
print(f"— the sum of the updated ranking vector:\n{np.sum(v)}")
```

```

initial ranking vector:
[[[0.2 0.2 0.2 0.2 0.2]]]
reformulated link matrix:
[[0.03 0.03 0.03 0.08 0.03 ]
 [0.31333333 0.03 0.03 0.03 0.08 ]
 [0.03 0.455 0.03 0.03 0.03 ]
 [0.31333333 0.03 0.03 0.03 0. ]
 [0.31333333 0.455 0.08 0.03 0.03 ]]

update of the ranking vector:
[[0.2 0.25666667 0.115 0.08666667 0.34166667]]

— the sum of the updated ranking vector:
0.000000000000002

```

```
n_iter = 10
for i in range(n_iter):
    # print('update of the ranking vector:\n', v.T)
    # print('the sum of all components of the updated ranking vector:\n', np.sum(v))
    v = L_d.dot(v)

print(f"{n_iter}th update of the ranking vector:\n{v.T}\n")
print(f"— the sum of the {n_iter}th updated ranking vector:\n{np.sum(v)}\n")

print(f"Ranking vector (by eigendecomposition):\n{ranking_eig}")
```

```
@10th Update of the ranking vector:
[0.07320928 0.34767099 0.17835163 0.05074263 0.35002547]]

— the sum of the 10th updated ranking vector:
..000000000000000000000000

Ranking vector (by eigendecomposition):
0.06092206 0.10408069 0.06920369 0.04226125 0.14528808 0.57831623]
```